

Proposed Max Block Size (PMBS)

The Proposed Max Block Size (PMBS) field specifies the maximum data size (in bytes) of the block that the Initiator supports, exclusive of block header fields, such as a block counter.

Start Offset (STARTOFS)

The Start Offset (STARTOFS) field is an optional 32-bit/64-bit length that specifies the offset in bytes from start of the file from which the Sender would start the transfer. This allows large file transfers to be segmented into multiple bulk transfer sessions. The RC[STARTOFS] bit indicates whether this start offset is present in the message. The RC[WIDERANGE] bit defines the size of start offset quantity (32-bit or 64-bit). Receivers are not required to accept non-zero start offset transfers. Devices SHOULD make every attempt to support non-zero start offset. If a Responder cannot accept a given start offset, it SHALL reject the SendInit with a StatusReport(GeneralCode: FAILURE, ProtocolId: BDX, ProtocolCode: START_OFFSET_NOT_SUPPORTED). See Table 109, “BDX Status reports”.

Length (DEFLN)

The optional length (DEFLN) field specifies a predetermined definite length for the transfer. For a SendInit message, this field defines the number of bytes the Sender commits to sending to the Receiver. For a ReceiveInit message, this field defines the maximum number of bytes that the Receiver wishes to receive from the Sender. The SendAccept or ReceiveAccept response will then contain the expected length for the transfer. A Receiver receiving a premature BlockEOF would have to consider the transfer failed. A length of 0 or a missing length field signifies an indefinite length. The RC(DEFLN) bit indicates the presence of this field. The RC[WIDERANGE] bit indicates the width of this field.

File Designator Length (FDL)

The File Designator Length (FDL) field is a 16-bit unsigned little-endian field over 2 octets that specifies the length of the upcoming file designator (FD) field, which has variable length.

File Designator (FD)

The File Designator (FD) field is a variable-length identifier chosen by the Initiator to identify the payload to be transferred. In some applications, this identifier will need to be negotiated in advance, so that the Responder will know how to handle the file designator. In other applications, the file designator and optional metadata will be sufficient for the Responder to determine whether to accept the file transfer and how to handle the data, allowing the transfer to proceed without any additional message exchanges.

The length of this field in bytes is provided by the previous File Designator Length (FDL) field.

Metadata (MDATA)

The Metadata (MDATA) field is an optional field, and allows the Initiator to send additional application-specific information about the file to be transferred. The contents of this field are not specified here; applications can use it to avoid needing a separate round-trip negotiation of the file designator, as described above. The TLV metadata consumes the rest of the payload for the SendInit/ReceiveInit message, after all previous fields.